SOLUTION
SECTION-I

Single Correct Answer Type **4 Q. [3 M (–1)]**

1. Ans. (A)

Sol. $J = \sigma E$ $\qquad \sigma = \frac{1}{\rho}$

$$J = \frac{E}{\rho} \Rightarrow E = \rho J \Rightarrow E \propto \rho$$

$E_1 : E_2 : E_3$

$\rho_1 : \rho_2 : \rho_3$

$4\rho : 2\rho : 6\rho$

$2 : 1 : 3$

2. Ans. (C)

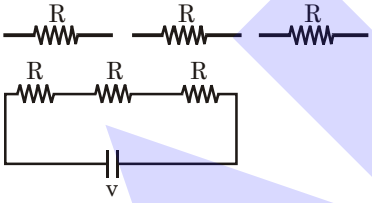
Sol. Apply Gauss's law : $\frac{q_{in}}{\varepsilon_0}$ = (outgoing flux - incoming flux)

$$V = IR = \frac{I\rho\ell}{A} \Rightarrow \frac{V}{\ell}A = I\rho \Rightarrow EA - I\rho$$

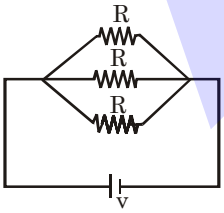
$$\Rightarrow \frac{q_{in}}{\varepsilon_0} = I\left|\rho_1 - \rho_2\right| \Rightarrow q_{in} = I\varepsilon_0\left|\rho_1 - \rho_2\right|$$

3. Ans. (D)

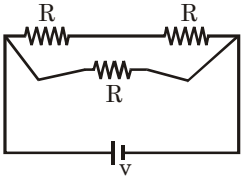
Sol.



Power dissipated in circuit = $\frac{v^2}{3R}$



Power dissipated in circuit = $\frac{v^2}{R/3}$



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Power dissipated in circuit = $\frac{v^2}{2R/3}$

4. Ans. (B)

Multiple Correct Answer Type **8 Q. [4 M (–1)]**

5. Ans. (B,C,D)

Sol. Current through the wire is same $i_A = i_B = i_C$ $\because$ i=neAV$_d$ and $J = \frac{i}{A}$, J = σE

6. Ans. (A, C)

Sol. $I = \frac{dV}{dR} = \frac{Edr}{\frac{dr}{\sigma 4\pi r^2}} = kE^2 4\pi r^2$ ..... (i)

$dV = E.dr$

from (1) $E^2 = \frac{I}{4\pi k r^2}$

$\Rightarrow dV = \int_a^b \sqrt{\frac{I}{4\pi k}} \frac{1}{r} dr = \sqrt{\frac{I}{4\pi k}} \ell n\left(\frac{b}{a}\right)$

7. Ans. (A,B,C)

8. Ans. (A,B,C)

Sol. Potential difference across resistance = Potential difference across the terminals of the battery.

So V=E – Ir

This is an equation of a straight line. Comparing this with the given graph, we can see that E = 10V and r = 5 Ω.

Also $I_{max} = \frac{E}{r} = \frac{10}{5} = 2A$ when external resistance R=0.

9. Ans. (C,D)

10. Ans. (A,B,D)

11. Ans. (B,C)

Sol. Heat produced = $\int I^2 R dt = 10\,[\,64 \times 1 + 16 \times 1 + 4 \times 1 + 1 \times 1] = 850$ J

Total charge = $\int Idt = [8+4+2+1] \times 1 = 15C$

12. Ans. (B, D)

Sol. $i = \frac{10}{2+8} = 1A$

$\int \left(\vec{E}_C + \vec{E}_{N.C}\right).\overrightarrow{d\ell} = iR$

Across 8Ω $E_{N.C} = 0$

$\int \vec{E}_C.\overrightarrow{d\ell} = 8$ J/C

(1) $\int \vec{E}_C.\overrightarrow{d\ell} = 8$ J

Since $\int_{loop} \vec{E}_C.\overrightarrow{d\ell} = 0$

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So work done by conservative electric field while an unit charge goes from Q to P is –8 joule.
Now across the battery, we have

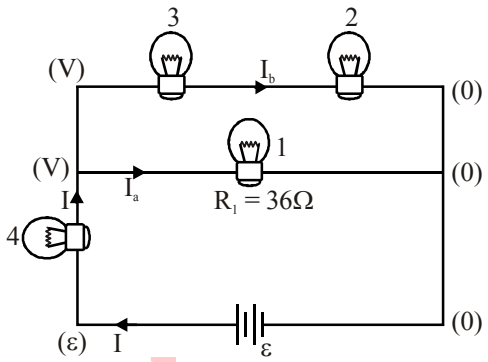
(1) $\int\left(\vec{E}_{N.C}\right).\overrightarrow{d\ell} = \varepsilon = 10\,\text{joule}$

$\therefore$ (B), (D)

Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

13. Ans. (B)
14. Ans. (A)
15. Ans. (B)

Sol. $i_2 = i_3 = I_b$



$V_1 = (V_2 + V_3) = V$
and $P_2 = P_3$
$\Rightarrow$ $R_2 = R_3$
and $V_2 = V_3 = V/2$
$P_1 = P_2 = P_3 = P_4$ (given)

$$P_1 = \frac{V^2}{36} \; ; P_3 = \frac{(V/2)^2}{R_3}$$

As $P_1 = P_3 \Rightarrow R_3 = 9\Omega$
Also $R_3 = R_2 = 9\Omega$ **Ans.**

$$I_a = \frac{I}{3} \text{ and } I_4 = I$$

$P_1 = P_4 = 4W$

$$\left(\frac{I}{3}\right)^2 R_1 = (I)^2 R_4 = 4W$$

$$\Rightarrow \quad \frac{R_1}{9} = R_4$$

$\Rightarrow$ $4\Omega = R_4$ **Ans.**
Also I = 1A
$R_{eq} = 16\Omega$ and I = 1A
$\varepsilon = 16V$

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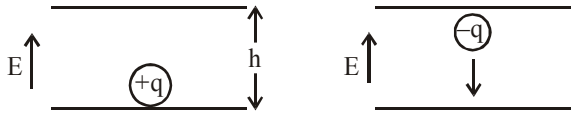
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16. Ans. (A)

Sol. Balls placed on +ve plate become positive charge and move upward due to electric field.
These balls on colliding with negative plate become negatively charged and move opposite to the direction of electric field.

17. Ans. (B)

Sol.



$$h = \frac{1}{2}at^2 \qquad [\text{as } u = 0]$$

$$\sqrt{\frac{2hm}{qE}} = \text{time} \Rightarrow \text{time} = \sqrt{\frac{2m}{q\Delta V}}$$

$$E = \frac{V_0}{h}$$

$$<\text{current}> = \frac{\text{charge}}{\text{time}} = \frac{q\sqrt{qV_0}}{\sqrt{2mh^2}}$$

$$q \propto V_0$$

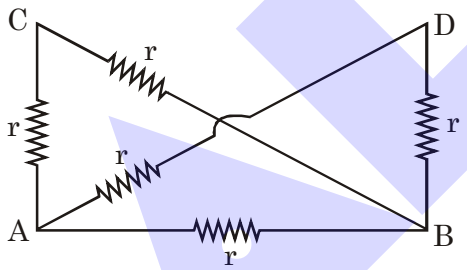
$$< I > \propto V_0^2$$

Linked Comprehension Type (1 Para × 3 Q.) [4 M (–1)]
(Multiple Correct Answer Type)

18. Ans. (A,D)

Sol. By symmetry

$V_C = V_D$



$$R_{AB} = \frac{r}{2}$$

19. Ans. (A,B,C,D)

20. Ans. (A,B,C)

SECTION-II

Numerical Answer Type Question 4Q.[3M(0)]
(upto second decimal place)

21. Ans. 12.00

Sol. Power developed in it is maximum when external resistance = internal resistance.

$mn = 324$

$$m = \frac{324}{n}, \; R_{eq} = \frac{nr}{m}$$

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$$\frac{nr}{324/n} = R \Rightarrow \frac{9n^2}{324} = 4 \Rightarrow n = 12$$

22. Ans. 25.00

Sol. Current in 60Ω resistance = 1A

$\therefore Q = i^2Rt = (1)^2 \times 60 \times 7 \times 60$ J

$\therefore Q = mS\Delta T \Rightarrow \Delta T = 25°C$

23. Ans. 20.00

Sol. $\left(\frac{V}{R+r}\right)^2 R = \left(\frac{V}{r+nR}\right)^2 \times nR$

$\Rightarrow r^2 + n^2R^2 = nR^2 + nr^2$

$(n^2 - n)R^2 = (n-1)r^2$

$nR^2 = r^2$

$r = \sqrt{n}\ R$

$r = 20\ \Omega$

24. Ans. 60.00

Sol. In parallel

$$H = \frac{V^2}{R/2}t_1 = \frac{2V^2}{R}t_1$$

In series

$$H = \frac{V^2}{2R}t_2$$

$$2t_1 = \frac{t_2}{2}$$

$t_2 = 4t_1 = 60$ min

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SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **1 Q. [4 M (0)]**

25. Ans. 2

Sol. $P_i = \left(\frac{V}{5+20}\right)^2 \times 5$

$$P_f = \left[\left(\frac{V}{\frac{5R}{5+R}+20}\right) \times \frac{5R}{5+R}\right]^2 \frac{1}{5}$$

$\Rightarrow P_f = \frac{P_i}{9}$

Subjective Type **5 Q. [4 M (0)]**

26. Ans. $T - T_0 = \frac{I^2R}{2\pi r\ell k}\left(1 - e^{-\frac{2kt}{rCd}}\right)$

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Sol. $\frac{dQ}{dt} = I^2R - kA(T - T_0)$

$$ms\frac{dT}{dt} = I^2R - k(2\pi r\ell)(T - T_0)$$

$$d\pi r^2\ell C\frac{dT}{dt} = I^2R - 2\pi r\ell k(T - T_0)$$

$$\int_{T_0}^{T}\frac{dT}{I^2R - 2\pi r\ell k(T - T_0)} = \int_0^t\frac{dt}{Cd\pi r^2\ell}$$

$$-\frac{1}{2\pi r\ell k}\ell n\left(\frac{I^2R - 2\pi r\ell k(T - T_0)}{I^2R}\right) = \frac{t}{Cd\pi r^2\ell}$$

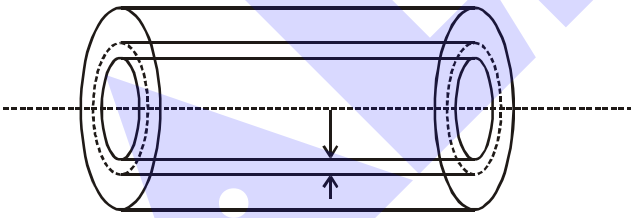
$$\ell n\left(\frac{I^2R - 2\pi r\ell k(T - T_0)}{I^2R}\right) = -\frac{2kt}{rCd}$$

$$1 - \frac{2\pi r\ell k}{I^2R}(T - T_0) = e^{-\frac{2kt}{rCd}}$$

$$\left[T - T_0 = \frac{I^2R}{2\pi r\ell k}\left(1 - e^{-\frac{2kt}{rCd}}\right)\right]$$

27. Ans. (a) $\frac{\Delta V}{L}, \frac{\Delta V r^2}{La}$; (b) $\frac{(\Delta V)^2(2\pi)}{La}\left(\frac{R^4}{4}\right)$, (c) $T_S = T_A + \frac{2\pi(\Delta V)^2}{La\lambda}\left(\frac{R^4}{4}\right)$, (d) $T = T_s + \frac{(\Delta V)^2}{16L^2aK}(R^4 - r^4)$

Sol.



$$E = \frac{\Delta V}{L},$$

$$J = \sigma E$$

$$= \frac{r^2}{a}\frac{\Delta V}{L}$$

$$P = i^2R = i^2\left(\frac{\rho L}{A}\right)$$

$$P = \int\left(\frac{\Delta V}{L}\frac{r^2}{a}\right)^2(2\pi r dr)^2.\left(\frac{a}{r^2}\frac{L}{2\pi r dr}\right)$$

$$P = \frac{(\Delta V)^2}{La}.2\pi\frac{R^4}{4}$$

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$P = \lambda(T - T_S)$

$$\frac{(\Delta V)^2}{La}.2\pi\frac{R^4}{4} = \lambda(T - T_S)$$

$$T = T_S + \frac{2\pi(\Delta V)^2}{La\lambda}.\frac{R^4}{4}$$

Also at a distance r from central axis

$$\frac{(\Delta V)^2}{La}.\frac{2\pi r^4}{4} = k(2\pi rL)\frac{dT}{dr}$$

$$\int dT = \frac{(\Delta V)^2}{4L^2ak}\int_r^R r^3 dr$$

$$T = T_s + \frac{(\Delta V)^2}{16L^2aK}(R^4 - r^4)$$

28. Ans. $\frac{1}{R} = \frac{\pi r^2}{3l}(2\sigma_2 + \sigma_1)$

29. **Ans.** (i) $\frac{1}{2}i_0t_0$ (ii) $i = i_0\left(1 - \frac{t}{t_0}\right)$ (iii) $\frac{Rt_0i_0^2}{3}$

Sol. (i) Total charge = Area under the i-t graph.

$$= \frac{1}{2}i_0t_0$$

(ii) $\frac{i}{i_0} + \frac{t}{t_0} = 1 \Rightarrow i = i_0\left(1 - \frac{t}{t_0}\right)$

(iii) $H = \int i^2R\,dt$

$$= \int_0^{t_0} i_0^2\left(1 - \frac{t}{t_0}\right)^2 \cdot R\,dt$$

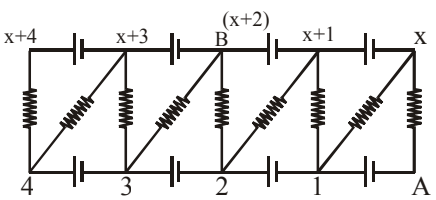
30. **Ans.** $-\frac{22}{9}V$

Sol. $$\frac{x+4-4}{1} + \frac{(x+3-4)}{1} + \frac{x}{1} + \frac{(x+2-3)}{1} + \frac{x}{1} + \frac{x+1-2}{1} + \frac{x}{1} + \frac{x-1}{1} + \frac{x}{1} = 0$$

$5x + 4(x - 1) = 0$

$9x - 4 = 0$

$x = \frac{4}{9}V$



$V_A - V_B = -(x + 2 - 0) = -\frac{22}{9}V$

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